TDNN and HMMs

agenda

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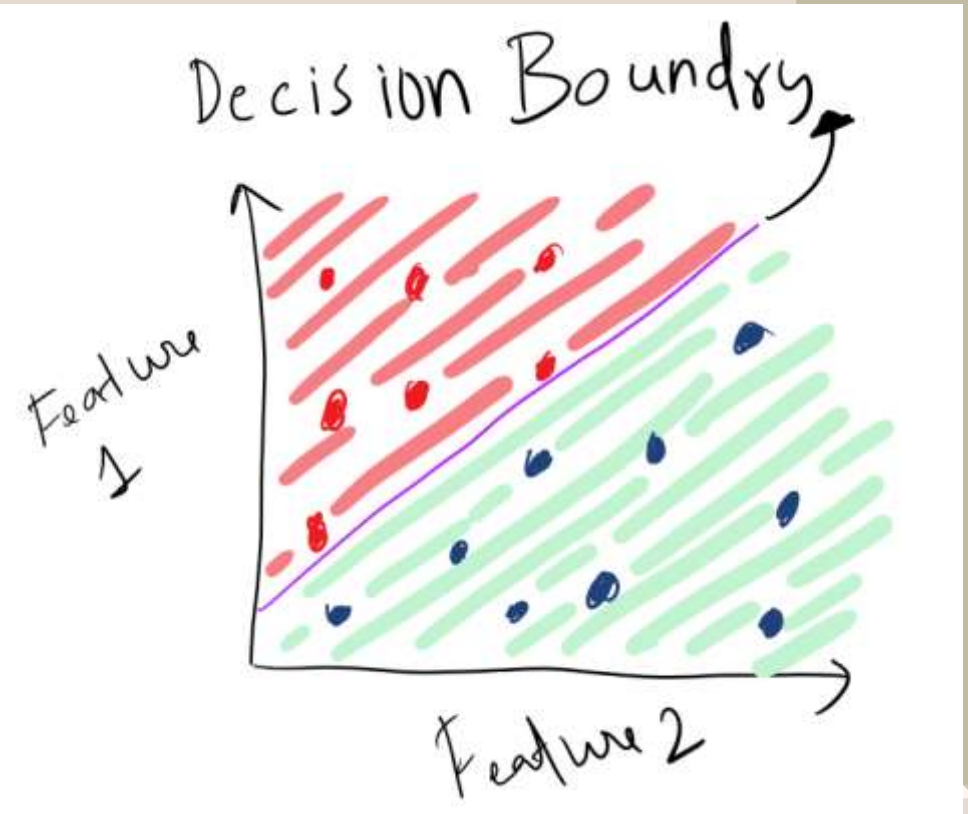
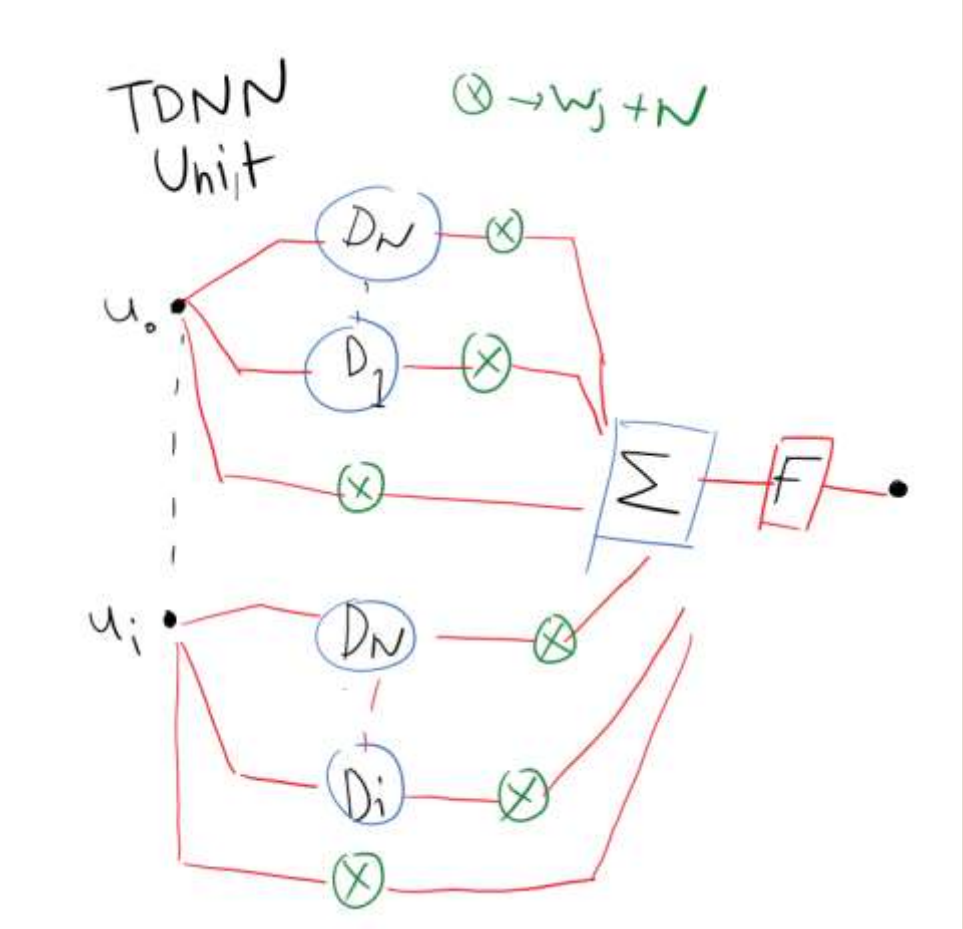
Introduction

- TDNNs was used to approach phenome recognition.
- Uses a 3 layer arrangement of simple computing units.
- The time delay arrangement enables the network to discover acoustic-phonetic features.
- TDNNs > HMMs. Other speech recognition models don’t come close.
- Neural Networks do give mixed results at times, however.



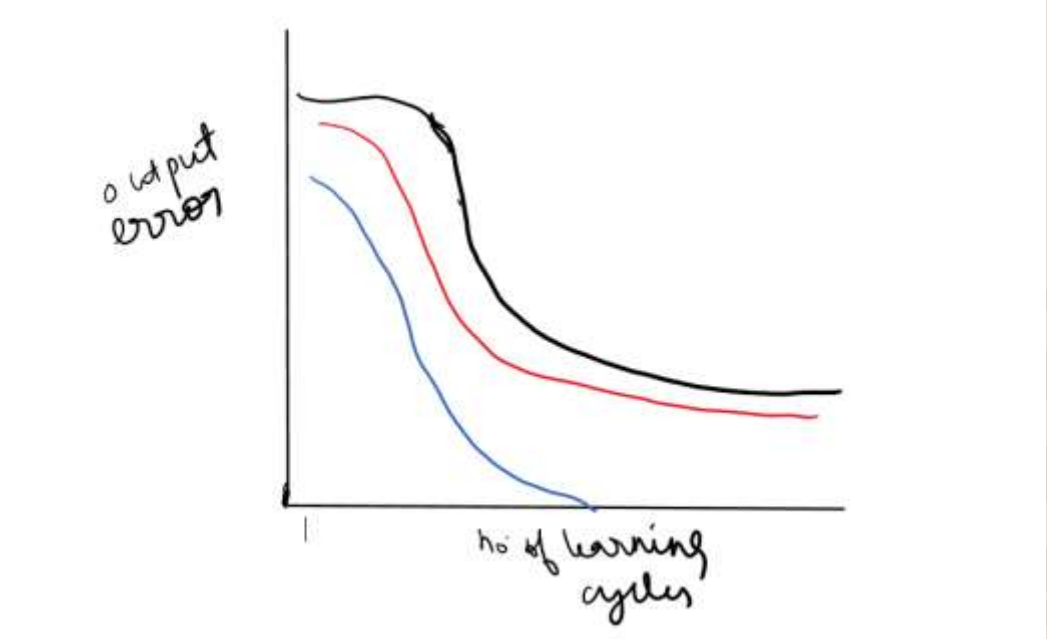
Time Delayed Neural Networks

- Should have multiple layers and sufficient interactions.
- Should be able to represent relationships between events in time.
- Actual features should be invariant under passage of time.
- Learning procedure should not require precise temporal alignment of labels that are to be learned.
- Number of weights should be small compared to data.



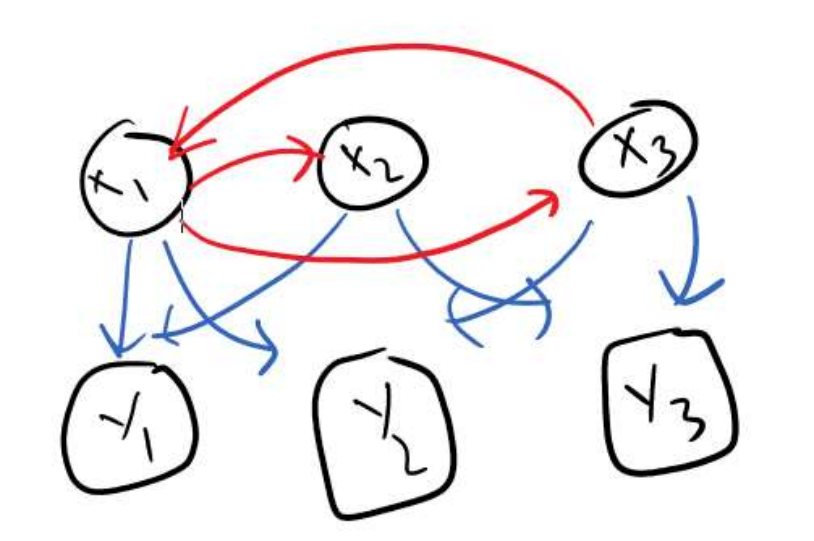
Learning in a TDNN

- It uses back propagation learning procedure. Adjusts all weights to decrease error in output.
- Since all the units are searching for same acoustic event in time, weights also should be same.
- Let there be 2 moments in time A and B.
- To keep weights synchronized, run all forward and back passes as if units were independent.



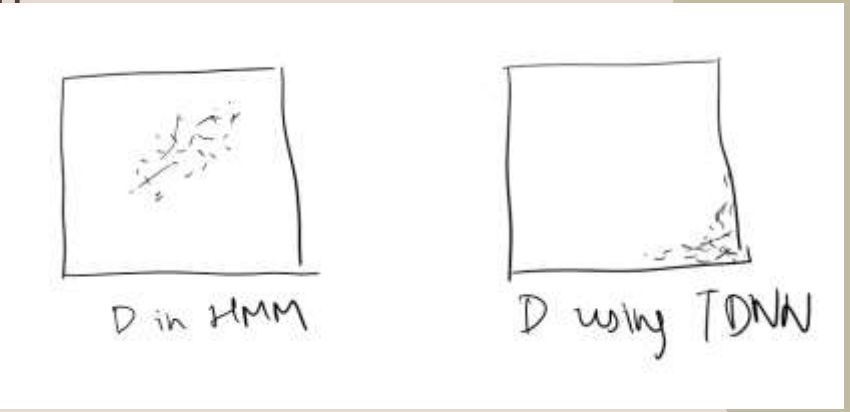
Hidden Markov Models

- Partially successful because they can deal with the variability in speech by means of stochastic modelling.
- Acoustic front end of HMM is usually vector quantizer that classifies short time spectra's.
- The Weighted Likelihood Ratio augmented with power values (PWLR) was used as LPC distance measure for vector quantization.
- Floor values were set on output probabilities to avoid errors caused by zero probabilities.



Recognition Experiments

- No preselection of tokens was done. Constants were taken from entire utterances and not in isolation.
- Variability exists in audio extracted.
- TDNN has tendency to fire with large confidence.
- Network has learnt to link variations in sensory input to same higher level concepts.
- TDNN and HMM after training are able to do one thing correctly based on sequences provided.



Conclusion

- TDNN invented abstractions in time and frequency like format tracking ang segmentation on its own.
- Links different acoustic events with same higher level concepts.
- Is translation invariant and does not rely on precise alignment or segmentation.

Thank You

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